Data Session 4 Key, part 2: Where the Votes Have to Come From

84-355 Democracy's Data | Session 4 — Returns over time (self-study) | Instructor copy — do not distribute

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August 11, 2026

All Part 5 code is given to students. Grade the interpretation.

Note the timing. This lab runs in **Session 4**, the self-study week (Sep 15/17), before Chapter 5 explains vote targeting. That is deliberate — Ch. 3 is about campaigns becoming professional, and this is what professional means — but it does mean students meet the method before the book's account of it. Expect some to say so; that is a fair observation and worth a point, not a deduction.

Every number below was run against the data file.

The lab in one sentence

Vote targeting turns one statewide number into a hundred county numbers by assuming the past distribution of your vote still holds — and 2024 is sitting in the data to show you what happens when it does not.

The findings

```
st <- aggregate(cbind(dem, rep, oth) ~ year, nc, sum); st
```

```
#>   year     dem     rep    oth
#> 1 2012 1931580 2440707  96008
#> 2 2016 2309157 2298880 102977
#> 3 2020 2834790 2586604  81383
#> 4 2024 3069496 2241309 280742
```

```
round(c(top10 = 100*sum(ar[1:10]), top25 = 100*sum(ar[1:25]),
        bottom50 = 100*sum(ar[51:100])), 1)
```

```
#>    top10    top25 bottom50
#>     37.5     63.1     15.0
```

- **Republican votes are concentrated but not extremely so**: the top 10 counties supply 37.5%, the top 25 supply 63.1%, and it takes **16 counties** to reach half the party's vote.
- **The book's 3.20M target is unreachable on this evidence.** The best Republican performance in the data is **2,586,604 (2020)**. The target is **613,396 above that**, and 958,691 above 2024.
- **2024 is a candidate-specific collapse, not a trend.** The Republican share falls to 40.1% and the "other" column runs at 3.4× its 2020 level.

The idea to teach

The arithmetic is trivial and completely correct. What makes it a lab is that **every step is an assumption wearing a number's clothing**. Averaging four elections assumes they are exchangeable. Using share-of-party-vote assumes the county's *relative* contribution is the stable thing. Picking 3.20M assumes a performance nobody has achieved.

Students who say "the method is fine, the inputs are guesses" have it.

Option A — A different statewide goal

Verified output

```r
# the three targets the lab offers: the book's 2028 goal, the winning
# Democratic total in 2024, and the best Republican total in the data
for (T in c(TARGET, DEM24, BEST)) {
  t <- ar * T
  cat(sprintf("target %9s | Wake %6.0f  Meck %6.0f  Durham %5.0f | top-5 order same: %s\n",
      format(T, big.mark=","), t["WAKE"], t["MECKLENBURG"], t["DURHAM"],
      identical(names(sort(t, decreasing=TRUE))[1:5], names(ar)[1:5])))
}
```

```
#> target 3,200,000 | Wake 267327  Meck 227753  Durham 40620 | top-5 order same: TRUE
#> target 3,069,496 | Wake 256425  Meck 218465  Durham 38963 | top-5 order same: TRUE
#> target 2,586,604 | Wake 216084  Meck 184096  Durham 32833 | top-5 order same: TRUE
```

target	Wake	Mecklenburg	Durham
3,200,000 (book's 2028 goal)	267,327	227,753	40,620
3,069,496 (2024 Democratic total)	256,425	218,465	38,963
2,586,604 (best Republican total, 2020)	216,084	184,096	32,833

The ranking never changes. Only the magnitudes scale.

What a good answer says

3/4: Reports the new targets and notices every county moves proportionally.

4/4: Draws the conclusion the prompt is fishing for — **changing the target does not change where the campaign goes, only how hard it has to work everywhere.** The target sets ambition; the *shares* set geography, and the shares are computed from history. A campaign that wants a different map of priorities has to change the shares, which means arguing the past is not a good guide — and that is a political judgment, not an arithmetic one.

4/4, best: compares the target to 2024 actuals and sees how large the gap is per county — Wake short by 109,415, Mecklenburg by 96,392. Asking a county organiser to find 109,000 additional votes is not a plan, it is a wish. A student who notices that the method produces *numbers* rather than *strategies* has understood something real about how campaigns talk to themselves.

Option B — Target the other party

Verified output

```
rbind(REP = round(100*c(top10=sum(ar[1:10]), top25=sum(ar[1:25]), bottom50=sum(ar[51:100])),1),
      DEM = round(100*c(top10=sum(ad[1:10]), top25=sum(ad[1:25]), bottom50=sum(ad[51:100])),1))
```

```
#>     top10 top25 bottom50
#> REP  37.5  63.1       15
#> DEM  55.6  73.7       11
```

```
c(counties_to_half_REP = which(cumsum(ar) >= .5)[1],
  counties_to_half_DEM = which(cumsum(ad) >= .5)[1])
```

```
#> counties_to_half_REP.BRUNSWICK    counties_to_half_DEM.ORANGE
#>                             16                              8
```

	top 10	top 25	bottom 50	counties to reach half
Republican	37.5%	63.1%	15.0%	**16**
Democratic	55.6%	73.7%	11.0%	**8**

Democratic top six: Wake, Mecklenburg, Guilford, Durham, Forsyth, Buncombe. Republican top six: Wake, Mecklenburg, Guilford, Forsyth, **Union, Gaston**.

What a good answer says

3/4: Notices the Democratic vote is more concentrated.

4/4: Says what that implies operationally. **8 counties versus 16 is a different campaign**, not a different emphasis: fewer field offices, denser media buys, a smaller travel radius, and far more exposure to one bad night in one metro. The Republican coalition is more spread out, which is more expensive to organise and more robust to a local collapse.

4/4, best: notices the top three counties are *the same for both parties*, because big counties are big. The lists diverge at positions 4–6, where Democrats add Durham and Buncombe and Republicans add Union and Gaston. Then asks the right question: is a county worth visiting because it has many of your voters, or because it has many *persuadable* voters? This method cannot tell you, and that is Option C.

Option C — Share of votes cast, not share of party votes

Verified output

```
nc$pct <- nc$rep / nc$total
bp <- sort(tapply(nc$pct, nc$county, mean), decreasing = TRUE)
data.frame(by_contribution = names(ar)[1:5], by_percentage = names(bp)[1:5])
```

```
#>   by_contribution by_percentage
#> 1            WAKE        YADKIN
#> 2     MECKLENBURG      MITCHELL
#> 3        GUILFORD     ALEXANDER
#> 4         FORSYTH         AVERY
#> 5           UNION      RANDOLPH
```

```
length(intersect(names(ar)[1:5], names(bp)[1:5]))
```

```
#> [1] 0
```

by contribution	by Republican percentage
Wake	Yadkin
Mecklenburg	Mitchell
Guilford	Alexander
Forsyth	Avery
Union	Randolph

Zero overlap. Not “a different order” — a disjoint list.

What a good answer says

3/4: Notices the lists are completely different and says the two measures answer different questions.

4/4: Assigns each ranking to a decision. **Contribution tells you where the votes are; percentage tells you where you are liked.** If the job is to bank raw votes, go to Wake. If the job is to find a friendly audience, go to Yadkin — which is 73.6% Republican and casts 3.3% as many votes as Wake. A rural county at 73.6% Republican may deliver fewer additional votes than a 36.1% share of a metro.

4/4, best: connects this to the campaign-visits material. Campaigns visit big competitive states, not the states that like them most, and this is the same logic one level down. Also notices that neither ranking mentions **margin** or **persuadability** — both are about where your existing voters live, so both implicitly treat the election as a turnout problem rather than a persuasion problem. That is a real strategic assumption, and Chapter 5 makes it mostly without argument.

Option D — Their own question

Good versions: which counties moved most between 2020 and 2024 (the Robinson collapse is not uniform); whether third-party voting in 2024 is concentrated where Robinson underperformed; whether county rank order is stable across all four elections.

Warn against comparing raw vote counts across counties of wildly different size without normalising.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is treating the target as an *assumption* rather than a goal. Students who audit the inputs beat students who extend the arithmetic.
- **Reward anybody who checks the book.** The lab closes on two errors in Table 5.3; a student who verifies rather than takes my word for it is doing exactly what the course is for.

Anticipated questions

- *"Why average across four elections instead of using the most recent?"* — Because one election is noisy. But averaging assumes exchangeability, and 2024 was not exchangeable. There is no clean answer; there is a defensible choice you have to state. Good discussion, five minutes.
- *"Is the book wrong?"* — Table 5.3's 2016 statewide figure is wrong (transposed digits), and 2020 is off by one. The five printed counties are right. Nothing in the chapter's argument depends on either. Say all three of those things — the point is the habit of checking, not scoring on a textbook.
- *"Why the governor's race rather than president?"* — Because the chapter uses it, and because a state race isolates state-level strategy from presidential coattails. Also because 2024 gives us a natural experiment in candidate collapse.
- *"Where is turnout?"* — Table 5.1 is registration and turnout; we work in votes because targeting works in votes. A student who asks for turnout as a denominator is thinking well — that is the ratio that tells you whether a county is maxed out.

Connections

- `campaign-visits` (currently retired for lack of a slot) — same strategic question at the state level. Run them together if you reinstate it.
- **Session 2 (apportionment)** — both are about converting a total into parts by a rule. There the rule is law; here it is a campaign's own choice.
- **Session 10 (live returns)** — this is a baseline-versus-actual exercise done in advance; that one does it after the fact with real returns.

- **This week's Tuesday debate** (deliberation, across two centuries) — Ch. 3 asks whether the transformation of campaigns raised the volume of information while lowering its quality. Targeting is the mechanism: it decides who hears anything at all.
- **Session 5 (money, message, and attention)** — when Ch. 5 arrives, students will already have done its worked example. Refer back to it explicitly; the chapter reads differently once the arithmetic is in your hands.